Live system

sensor gateway
956 spaces × 3 sensors, ~400 readings/s
noise · drift · stuck faults

edge controllers (×6)
decisions · alerts · door events
heartbeats · if_ms

server agent
verdicts: confirm / watch / stand-down
llm_ms · takeover · failover

MQTT

MQTT

MQTT

TimescaleDB
sensor_readings · edge_decisions
heartbeats · llm_overrides · fire_events

Grafana
Operations · Technical

readings + verdicts
(verdicts are the labels)

fds.models.retrain (ClearML task)
1. per-room 5 s windows from real readings
2. + windows the LLM rejected (×20)
3. + augmented false alarms (smoke, no CO)
4. IsolationForest, contamination 0.02
5. evaluate vs. the deployed model
6. export ONNX, check sklearn/ONNX parity

detector.onnx

new model on every controller

scripts/deploy.sh edge
build arm64 + amd64 → import on each node
rollout restart of all six controllers